

# MID-SEMESTER EXAMINATION 2025-26 (ODD)

Prog: B.Tech

Subject :- PROGRAMMING FOR PROBLEM SOLVING

Branch: CSE & Allied

Sub. Code: CS101

Time: 1 Hour

Semester: 1<sup>st</sup>

GROUP-1,2,3,4,5,6,7,8,9, 10

Max marks 20

The figures in the right-hand margin indicate marks.

## PART A

- Q1. Answer ALL questions:
- What is Platform Dependency and why is C called Platform dependent Language? Explain with diagram.
  - Explain all the phases of C source code compilation and execution with all intermediate files, using a diagram.
  - Explain all the symbols used in a flowchart and their meanings.
  - Explain all integer data types with sizes and limits(ranges).
  - Write a program to check the input year is Leap year or Not.

## PART-B (Answer any two)

Q2. Write a Program and Flow Chart to check the given number is Prime or Not.

Q3. Write a program to display the following shape using Nested For Loops.

```

1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
    
```

Q4. Write a C program to count all prime numbers in a 2D array and display those elements along with their positions.

```

7   13   9   4
8   6   12  5
9   17  11  22
45  1   87  2
    
```

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Registration No.:

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Total number of printed pages – 3

B. Tech

Subject Code: CS101

FIRST SEMESTER REGULAR EXAMINATION – 2025  
SUBJECT NAME: Programming For Problem Solving  
BRANCH: CSE & ALLIED Branches  
Time – 3 Hours, Full Marks – 100  
Answer all Questions from PART – A & B

PART -A

Q1. Specify the outputs: 2 x 15 (Read the programs carefully)

|      |                                                                                                                                                         | CO | M<br>ar<br>ks | B<br>L |       |                                                                                                                                                    | C<br>O | M<br>ar<br>ks | BL |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------|--------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|----|
| i)   | <pre>#include &lt;stdio.h&gt; void main() {     int result = 6 + 2 * 3 - 4 / 2;     printf("Result is: %d\n",     result); }</pre> <p>10</p>            | 1  | 2             | 2      | ii)   | <pre>#include &lt;stdio.h&gt; void main() {     double num = 9.45;     int result = (int)num;     printf("Result: %d\n", result); }</pre> <p>9</p> | 1      | 2             | 2  |
| iii) | <pre>#include &lt;stdio.h&gt; void main() {     int i, sum=0;     for(i=-3; i &lt;= 3; i++)         if (i) { sum+=i};     printf("%d ", 6 / i); }</pre> | 1  | 2             | 2      | iv)   | <pre>#include &lt;stdio.h&gt; void main() {     int a = 2; int b = a+ ++a;     printf("%d", b); }</pre> <p>5</p> <p>a+(1+a)<br/>2+3</p>            | 2      | 2             | 2  |
| v)   | <pre>#include &lt;stdio.h&gt; void main() {     int x,i,sum=0;     for(i=0;i&lt;50;i++);     sum=sum+i;     printf("%d", sum); }</pre>                  | 2  | 2             | 2      | vi)   | <pre>#include &lt;stdio.h&gt; void main() {     union abc {int x;float y; char     z[10];} X;      printf("%d", sizeof(X)); }</pre> <p>4</p>       | 2      | 2             | 2  |
| vii) | <pre>#include &lt;stdio.h&gt; void main(){     int x = 4, y=5; z;     z = (++x/y++);     printf("%d \t %d\t%d",     x,y,z); }</pre> <p>6, 5, 1</p>      | 3  | 2             | 2      | viii) | <pre>#include &lt;stdio.h&gt; void main() {     int i=3;     while(i&gt;0)     { printf("%d\t",i); i--;} }</pre> <p>3 3 2 1</p>                    | 3      | 2             | 2  |

3, 3, 2, 1

3, 2, 1

1

1+2+3+4+5+6+7+8+9  
= 45

3,

370  
{ 80 ( "0/d 14", i);  
i--;



|       |                                                                                                                                                                                                          |   |   |   |      |                                                                                                                                                                                                                 |   |   |   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| ix)   | <pre>#include&lt;stdio.h&gt; void main() { int i=1; do { printf("%d\t",i); i+=3; } while(i&lt;10); }</pre> <p>147</p>                                                                                    | 3 | 2 | 2 | x)   | <pre>#include &lt;stdio.h&gt; void main () { int i=1, sum=0; while (i&lt;=10 &amp;&amp; i++) { sum=i++;} printf("\n%d", sum); }</pre>                                                                           | 3 | 2 | 2 |
| xi)   | <pre>#include&lt;stdio.h&gt; void main() { int x=5, *p=&amp;x; *p=8; printf("%d",++x); }</pre> <p>a</p>                                                                                                  | 4 | 2 | 2 | xii) | <pre>#include &lt;stdio.h&gt; void main() { int a[5][5]={1,2,3},{4,5,6,7},{8},{9,10} }; for (int i=0; i&lt;5; i++) { printf("\n"); for (int j=0; j&lt;5; j++) printf("%d\t", a[i][j]); } }</pre>                | 4 | 2 | 3 |
| xiii) | <pre>#include&lt;stdio.h&gt; void main() { struct abc { int x;float y; char z;} str; printf("%d", sizeof(str)); }</pre> <p>Assume integer is 4 bytes, Float is 4 bytes, Character is 1 byte</p> <p>a</p> | 5 | 2 | 2 | xiv) | <pre>#include&lt;stdio.h&gt; void main() { struct abc {int x;float y; char z;}; struct abc str[10]={{1,2.0,'A'}, {3,5.0,'B'}}; printf("%d %.2f", (str[0].x+str[2].x), (str[0].y+str[2].y)); }</pre> <p>2.00</p> | 5 | 2 | 2 |
| xv)   | <pre>#include&lt;stdio.h&gt; void main() { int a[10]= {1,2,3,4,5,6,7,8,9,10}; int sum=0; for (int i=0; i&lt;10; i+=2) sum+=*(a+i); printf("%d", sum); }</pre> <p>1491625</p>                             | 5 | 2 | 2 |      |                                                                                                                                                                                                                 |   |   |   |

| PART-B |                                                                                                                                                                                                                               | CO | Marks | BL |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|----|
| Q2.    | Answer any two:                                                                                                                                                                                                               |    |       |    |
| (a)    | (i) Write a C program to find factorial of a given integer using recursion.<br><br>(ii) Write a C program to input your Date of Birth and the current date in DD/MM/YYYY format and find your age in YEARS, MONTHS, and DAYS. | 1  | 7     | 2  |
| (b)    | (i) Write a C program to check if a given integer is Prime or not.<br><br>(ii) Write a C program to find sum of odd integers and even integer between M and N.                                                                | 1  | 7     | 2  |
| (c)    | Write a menu driven C program that reads two integers and performs all arithmetic operations on them. Display the results of addition, subtraction, multiplication, division and modulus. (use functions).                    | 1  | 7     | 2  |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|
| <b>Q3.</b> | <b>Answer any two:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |    |   |
| (a)        | Write a C program to input marks of a student and display the grade using if else.<br>Use the following grade rules:<br><ul style="list-style-type: none"> <li>• 90 to 100 gives O grade</li> <li>• 80 to 89 gives E grade</li> <li>• 70 to 79 gives A grade</li> <li>• 60 to 69 gives B grade</li> <li>• Below 60 gives F</li> </ul>                                                                                                                                                                                         | 2 | 7  | 3 |
| (b)        | Write a C program that accepts two 3×3 matrices from the user and computes their product. The program should correctly apply matrix multiplication rules. After computing the product, display the resulting 3×3 matrix in a properly formatted manner.                                                                                                                                                                                                                                                                       | 2 | 7  | 3 |
| (c)        | Create a menu-based C program using switch-case that calculates the area of different shapes. Provide options for Circle, Rectangle, Triangle. The program should read the required inputs for each choice and display the area.                                                                                                                                                                                                                                                                                              | 2 | 7  | 3 |
| <b>Q4.</b> | <b>Answer any two:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |    |   |
| (a)        | Write a C program to concatenate 2 strings [String1= "I LOVE", String2="CGU"] without using the strcat () function. Also count how many vowels are in the concatenated string.                                                                                                                                                                                                                                                                                                                                                | 3 | 7  | 3 |
| (b)        | Write a C program to create an array of 10 employees with details: <b>employee ID, Name, Date of Birth, Age</b> . Find the name of the oldest employee and youngest employee.                                                                                                                                                                                                                                                                                                                                                 | 3 | 7  | 3 |
| (c)        | Write a program to create an array of 10 integers and find their sum and average by using pointers.                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 | 7  | 3 |
| <b>Q5.</b> | <b>Answer all the question.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |   |
| (a)        | (i) Write a C program to swap the values of two integer variables using a function and print the values before and after swapping.<br><br>(ii) Write a C program that uses a user-defined function to compute the square of each number from 1 to N.<br><br>(iii) Write a C program to evaluate ${}^nC_r$ (combination) and ${}^nP_r$ (permutation) for two given values of n and r using function(s). Validate the values of n and r before.<br><br>(iv) Write a C program to find the sum of the digits of a given integer. | 4 | 14 | 2 |
| <b>Q6.</b> | <b>Answer all the question.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |   |
|            | (i) Write down the different modes of accessing a file in C.<br><br>(ii) CGU University wants to maintain a database of 200 students containing fields such as <b>student name, registration number, age, department, address, date of admission and semester</b> . Write a C program to store all these records into a file named CGU_STUDENTS. Also find the average age of the students.                                                                                                                                   | 5 | 14 | 2 |

Best of Luck